MATHEMATICS / INDEPENDENT STUDY

MM7033

Advanced Algebra

Complete solutions

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

1 Unit 1 · Fields and splitting fields

MM7033-U1.01 · E · 15 min
Learning objective: U1.1 Minimal polynomials
Find the minimal polynomial and degree of $\sqrt{11}$ over $\mathbb{Q}$.
Decisive idea. Eisenstein applies at eleven.
$x^2 - 11$ is monic, irreducible by Eisenstein, and vanishes at the element. It is the minimal polynomial, so the degree is two.
Check your argument. Verify all three properties.

MM7033-U1.02 · E · 15 min
Learning objective: U1.1 Splitting fields
Find a splitting field of $x^2 - 3$ over $\mathbb{Q}$.
Decisive idea. Adjoining either root supplies the other.
The field $\mathbb{Q}\left(\sqrt{3}\right)$ contains both roots $\pm\sqrt{3}$ and is generated by them, so it is a splitting field. It has degree two because the polynomial is irreducible.
Check your argument. State the generation condition as well as splitting.

MM7033-U1.03 · E · 15 min
Learning objective: U1.2 Finite-field subfields
List the subfield sizes in $\mathbb{F}_{3^4}$.
Decisive idea. The degrees are one, two, and four.
The subfields have sizes $3, 9, 81$, each unique. No other degree can occur by the tower law, and the finite-field subfield theorem supplies these three.
Check your argument. Include uniqueness and exhaustiveness.

MM7033-U1.04 · E · 15 min
Learning objective: U1.2 Separability
Show $x^5 - x$ over $\mathbb{F}_5$ has no repeated roots.
Decisive idea. It is negative one.
The derivative is $5x^4 - 1 = -1$, so its gcd with the polynomial is one. Thus the polynomial is separable. It in fact has all five field elements as roots.
Check your argument. Use the derivative criterion, with the correct characteristic.

MM7033-U1.05 · C · 30 min
Learning objective: U1.1 Splitting-field degree

Find the degree of the splitting field of $x^3 - 3$ over $\mathbb{Q}$.

Decisive idea. The first degree is three, and the second is two because the first field is real.

Eisenstein gives $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ for the real root α. The other roots are $\alpha\zeta, \alpha\zeta^2$, so the splitting field is $\mathbb{Q}(\alpha, \zeta)$. Since $\mathbb{Q}(\alpha)$ is real, it does not contain nonreal ζ, whose quadratic polynomial is $x^2 + x + 1$. The total degree is six.

Check your argument. Justify both steps of the tower.

MM7033-U1.06 · C · 30 min
Learning objective: U1.1 Embeddings

Describe all $\mathbb{Q}$-embeddings of $\mathbb{Q}\left(\sqrt{2}\right)$ into $\mathbb{C}$.

Decisive idea. There are two choices of root.

The generator can map only to $\sqrt{2}$ or $-\sqrt{2}$. Each choice extends uniquely to an embedding by the root-extension theorem, with formula $a + b\sqrt{2} \mapsto a \pm b\sqrt{2}$. Hence there are exactly two.

Check your argument. Show the possible images actually define embeddings.

MM7033-U1.07 · C · 30 min
Learning objective: U1.2 Frobenius order

Prove Frobenius on $\mathbb{F}_{p^n}$ has order exactly n.

Decisive idea. A smaller positive power would make all elements roots of a polynomial of degree less than p^n.

The nth iterate is $a \mapsto a^{p^n} = a$. If the kth iterate were identity for $0 < k < n$, all p^n elements would solve $x^{p^k} - x = 0$, contradicting the root bound since $p^k < p^n$. Thus the order is exactly n.

Check your argument. Exclude every smaller positive exponent.

MM7033-U1.08 · C · 30 min
Learning objective: U1.2 Derivatives

Prove a polynomial over a field of characteristic p has derivative zero exactly when all nonzero terms have exponents divisible by p.

Decisive idea. The coefficient of x^{i-1} is ia_i.

The derivative is zero precisely when each $ia_i = 0$. If $a_i \neq 0$, the field property makes this equivalent to $i = 0$ in the prime field, meaning $p|i$. The converse follows by differentiating such monomials.

Check your argument. Use uniqueness of polynomial coefficients.

MM7033-U1.09 · C · 30 min
Learning objective: U1.1 Changing the base

Compare the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ and over $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$.

Decisive idea. Its degree over that base is one.

Over $\mathbb{Q}$ the polynomial is $x^2 - 2$. Over the larger field it is $x - \sqrt{2}$. Minimal polynomials and degrees depend on the specified base, even when the element is unchanged.

Check your argument. State both coefficient fields explicitly.

MM7033-U1.10 · A · 50 min
Learning objective: U1.1 Quartic splitting field

Prove the splitting field of $x^4 - 2$ over $\mathbb{Q}$ has degree eight.

Decisive idea. Its roots are $\pm\alpha, \pm i\alpha$.

Let α be the positive real fourth root of two. Eisenstein at two gives degree four for $\frac{\mathbb{Q}(\alpha)}{\mathbb{Q}}$. The splitting field is $\mathbb{Q}(\alpha, i)$, since adjoining i supplies all four roots and the ratio of $i\alpha$ to α recovers i. The field $\mathbb{Q}(\alpha)$ is real, so $x^2 + 1$ has no root there and adjoining i has degree two. The tower degree is eight.

Check your argument. Prove the named field is generated by the roots and compute the relative quadratic degree.

MM7033-U1.11 · A · 50 min
Learning objective: U1.2 Subfield intersections

Inside $\mathbb{F}_{p^{12}}$, determine the intersection of its subfields of orders p^4 and p^6.

Decisive idea. Both subfields contain the unique field of order p^2.

The intersection is a subfield, so its degree over $\mathbb{F}_p$ divides four and six, hence divides two. Both larger subfields contain the unique order-p^2 subfield, because two divides both degrees and uniqueness identifies their copies inside the ambient field. Thus the intersection has degree at least two and therefore exactly two.

Check your argument. Prove both a lower and upper bound on the intersection.

MM7033-U1.12 · A · 50 min
Learning objective: U1.2 Inseparable irreducibles

Prove $x^p - t$ is irreducible and inseparable over $\mathbb{F}_{p(t)}$.

Decisive idea. Use Eisenstein at the prime element t, then the derivative.

It is primitive and Eisenstein at t: all nonleading coefficients are divisible by t and the constant term is not divisible by t^2. Gauss's lemma gives irreducibility over the fraction field. Its derivative in x is zero. In a splitting field with root α, the characteristic-p identity gives $x^p - t = (x - \alpha)^p$, so its root repeats and it is inseparable.

Check your argument. Explain why the finite-field separability theorem does not apply to the infinite coefficient field.

2 Unit 2 · Existence and transcendence

MM7033-U2.01 · E · 15 min
Learning objective: U2.1 Chains

Give a partially ordered set with two maximal elements and no greatest element.
Decisive idea. A two-element set with only equality as its order suffices.
Let the elements be a, b, with no comparison between distinct elements. Both are maximal because neither has a strictly larger element. Neither is greatest because it is not above the other.
Check your argument. Distinguish the definitions of maximal and greatest.

MM7033-U2.02 · E · 15 min
Learning objective: U2.1 Proper ideals

Prove an ideal I of a unital ring is proper exactly when $1 \notin I$.
Decisive idea. The converse follows because the whole ring contains one.
If $1 \in I$, then every $r = r1$ lies in I, so $I = R$. Conversely if $I = R$, one belongs to it. Taking the negation gives the criterion.
Check your argument. Use the ideal absorption property.

MM7033-U2.03 · E · 15 min
Learning objective: U2.2 Independence

Is $\{t, t^2\}$ algebraically independent over K in $K(t)$?
Decisive idea. Use $Y - X^2$.
No. The nonzero polynomial $Y - X^2 \in K[X, Y]$ evaluates to zero at (t, t^2). The singleton $\{t\}$ is independent by the definition of the indeterminate.
Check your argument. Exhibit a nonzero formal polynomial relation.

MM7033-U2.04 · E · 15 min
Learning objective: U2.2 Transcendence bases

Give a transcendence basis of an algebraic extension $\frac{L}{K}$.
Decisive idea. The empty set is independent.
The empty set is a transcendence basis: $K(\varnothing) = K$, and $\frac{L}{K}$ is algebraic by hypothesis. Independence of the empty set means a nonzero constant cannot vanish.
Check your argument. Check both parts of the definition.

MM7033-U2.05 · C · 30 min
Learning objective: U2.1 Chain unions

Prove the union of a nonempty chain of proper ideals is proper and an ideal.

Decisive idea. If one were in the union, it would be in one member.

For elements in the union, their two ideals are comparable, so both lie in the larger one; their difference belongs there. Multiplication by a ring element preserves whichever ideal contains the element. If one lay in the union, some member would contain one and be the whole ring, a contradiction.

Check your argument. The empty chain requires a separately chosen upper bound in a Zorn application.

MM7033-U2.06 · C · 30 min
Learning objective: U2.1 Maximal ideals over an ideal

Use maximal-ideal existence in a quotient to show every proper ideal I lies in a maximal ideal.

Decisive idea. Pull back a maximal ideal under the quotient map.

$\frac{R}{I}$ is a nonzero commutative unital ring, so it has a maximal ideal M'. Its inverse image M contains I, and $\frac{R}{M} \cong \frac{\frac{R}{I}}{M'}$ is a field. Therefore M is maximal.

Check your argument. Use the quotient criterion for maximality.

MM7033-U2.07 · C · 30 min
Learning objective: U2.2 Algebraic closure

Show an algebraically closed field has no proper algebraic field extension.

Decisive idea. An irreducible polynomial over an algebraically closed field is linear.

For algebraic α over K, its minimal polynomial splits over the algebraically closed K. Since it is irreducible, it must be linear, so $\alpha \in K$. Every element of an algebraic extension therefore already belongs to K.

Check your argument. Apply the reasoning to every extension element.

MM7033-U2.08 · C · 30 min
Learning objective: U2.2 Transitivity

Prove algebraicity is transitive through a tower, allowing infinite extensions.

Decisive idea. Put those coefficients into a finite intermediate extension.

If α is algebraic over algebraic $\frac{L}{K}$, collect the finitely many coefficients of an annihilating polynomial into $F = K(c_1, \ldots, c_r)$. Then $\frac{F}{K}$ and $\frac{F(\alpha)}{F}$ are finite. The tower law makes $\frac{F(\alpha)}{K}$ finite, so α is algebraic over K.

Check your argument. Do not assume $\frac{L}{K}$ itself has finite degree.

MM7033-U2.09 · C · 30 min
Learning objective: U2.1 Extending independent sets

Use Zorn's lemma to extend a linearly independent subset of a vector space to a basis.

Decisive idea. A finite relation in a chain union lies in one chain member.

Order the linearly independent subsets containing the prescribed independent set S by inclusion. This poset is nonempty because it contains S. Every chain union remains independent by finite dependence and is an upper bound. Zorn gives a maximal independent superset. If it failed to span, adding a vector outside its span would preserve independence, contradicting maximality. Thus it is a basis.

Check your argument. Preserve the originally specified independent set throughout.

MM7033-U2.10 · A · 50 min
Learning objective: U2.2 A nontrivial transcendence basis

Show $\{t^2\}$ is a transcendence basis of $\frac{K(t)}{K}$.

Decisive idea. The element t satisfies $X^2 - t^2$ over $K(t^2)$.

If a nonzero polynomial P vanished at t^2, then $P(X^2)$ would be a nonzero polynomial vanishing at the indeterminate t, impossible. Thus t^2 is transcendental. The element t is algebraic over $K(t^2)$ through $X^2 - t^2$, so $\frac{K(t)}{K(t^2)}$ is algebraic. These two facts prove the basis property, in every characteristic.

Check your argument. Do not assume separability in characteristic two.

MM7033-U2.11 · A · 50 min
Learning objective: U2.1 Units detected by maximal ideals

Prove an element of a nonzero commutative unital ring is a unit exactly when it belongs to no maximal ideal.

Decisive idea. Extend that ideal to a maximal ideal.

A unit cannot lie in a proper ideal, hence in no maximal ideal. Conversely if r is not a unit, (r) is proper: an equation $1 = sr$ would make it a unit. Maximal-ideal existence places (r) inside a maximal ideal, which contains r. Taking the contrapositive gives the result.

Check your argument. Use commutativity for the principal-ideal/unit equivalence.

MM7033-U2.12 · A · 50 min
Learning objective: U2.2 Maximal independence

Prove every maximal algebraically independent subset B of $\frac{L}{K}$ is a transcendence basis.

Decisive idea. A relation after adjoining it would be a polynomial in that element with coefficients in $K(B)$.

Suppose α is transcendental over $K(B)$. A nonzero polynomial relation among finitely many elements of B and α can be grouped by powers of α. At least one coefficient polynomial in the distinct elements of B is nonzero after evaluation, by independence. Thus the relation makes α algebraic over $K(B)$, impossible. Hence $B \cup \{\alpha\}$ would be independent, contradicting maximality. Every element is therefore algebraic over $K(B)$.

Check your argument. Explain why the coefficient polynomial does not vanish identically after evaluation.

3 Unit 3 · Modules and quotient constructions

MM7033-U3.01 · E · 15 min
Learning objective: U3.1 Annihilators
Compute the annihilator of $[4]$ in the $\mathbb{Z}$-module $\mathbb{Z}/12\mathbb{Z}$.
Decisive idea. This holds exactly when three divides r.
The annihilator is $3\mathbb{Z}$, because $12|4r$ iff $3|r$. The cyclic submodule generated by $[4]$ is consequently isomorphic to $\mathbb{Z}/3\mathbb{Z}$.
Check your argument. Specify the element annihilator rather than the whole-module annihilator.

MM7033-U3.02 · E · 15 min
Learning objective: U3.1 Module maps
Describe all $\mathbb{Z}$-module maps $\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$.
Decisive idea. Any of the five residue classes can be chosen.
For each $a \in \mathbb{Z}/5\mathbb{Z}$, the map is $n \mapsto na$. These are additive and hence $\mathbb{Z}$-linear, and every map has this form by additivity. There are five.
Check your argument. Explain existence and exhaustiveness.

MM7033-U3.03 · E · 15 min
Learning objective: U3.2 Submodules
Show an ideal of R is exactly a submodule of R as an R-module.
Decisive idea. Both require an additive subgroup closed under multiplication by every scalar.
A submodule of the regular module is an additive subgroup with ra in it whenever $r \in R$ and a belongs. In a commutative ring this is exactly an ideal's absorption condition. Thus the notions coincide.
Check your argument. State the role of commutativity if discussing two-sided ideals.

MM7033-U3.04 · E · 15 min
Learning objective: U3.2 Quotients
Describe the quotient $\frac{\mathbb{Z}^2}{N}$ for $N = \{(a, 0) : a \in \mathbb{Z}\}$.
Decisive idea. The projection is onto with kernel N.
The map $(a, b) \mapsto b$ is a surjective module homomorphism with kernel N. Thus the quotient is isomorphic to $\mathbb{Z}$, by the first module isomorphism theorem.
Check your argument. Provide the map rather than an unsupported dimension argument.

MM7033-U3.05 · C · 30 min
Learning objective: U3.1 Cyclic homomorphisms

Compute the number of $\mathbb{Z}$-module maps $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$.

Decisive idea. Count residues a with $n|ma$.

A map is uniquely determined by $a = f([1])$ and is well defined exactly when $ma = 0$ modulo n. Writing $d = \gcd(m, n)$, this means a is a multiple of $\frac{n}{d}$ modulo n. There are exactly d such residues, hence d homomorphisms.

Check your argument. Prove the condition is sufficient as well as necessary.

MM7033-U3.06 · C · 30 min
Learning objective: U3.1 Operator modules

Let T on K^2 satisfy $T(e_1) = 0, T(e_2) = e_1$. Show its $K[x]$-module is cyclic and identify it as a quotient.

Decisive idea. The annihilator of e_2 is (x^2).

The vectors e_2 and $xe_2 = e_1$ span, so the module is cyclic. For $f = a_0 + a_1x + x^2q$, $f(T)e_2 = a_0e_2 + a_1e_1$, which vanishes exactly when $a_0 = a_1 = 0$. Thus the annihilator is (x^2) and the module is $\frac{K[x]}{x^2}$.

Check your argument. Compute the full annihilator, not only one annihilating polynomial.

MM7033-U3.07 · C · 30 min
Learning objective: U3.2 Isomorphism theorem

In $\mathbb{Z}$, take submodules $A = 4\mathbb{Z}$ and $B = 6\mathbb{Z}$. Identify $\frac{A+B}{B}$ and $\frac{A}{A\cap B}$.

Decisive idea. The sum is $2\mathbb{Z}$ and intersection is $12\mathbb{Z}$.

The two modules are $2\frac{\mathbb{Z}}{6}\mathbb{Z}$ and $4\frac{\mathbb{Z}}{12}\mathbb{Z}$, each cyclic of order three. The isomorphism sends the class of an element of $4\mathbb{Z}$ modulo $12\mathbb{Z}$ to its class modulo $6\mathbb{Z}$. Its kernel and image are exactly those in the second isomorphism theorem.

Check your argument. State the induced map and distinguish the ambient quotients.

MM7033-U3.08 · C · 30 min
Learning objective: U3.2 Simple modules

Classify the simple $\mathbb{Z}$-modules up to isomorphism.

Decisive idea. They are generated by primes.

Every simple module over a commutative ring is $\frac{R}{M}$ for a maximal ideal. The maximal ideals of $\mathbb{Z}$ are $p\mathbb{Z}$ with p prime. Thus the simple $\mathbb{Z}$-modules are exactly $\frac{\mathbb{Z}}{p}\mathbb{Z}$. Different primes give different cardinalities.

Check your argument. Exclude the zero module by the definition of simple.

MM7033-U3.09 · C · 30 min
Learning objective: U3.1 Torsion
For a domain R, prove the torsion elements of an R-module form a submodule.
Decisive idea. The product of the annihilators is still nonzero.
If $am = 0$ and $bn = 0$ with nonzero a, b, then $ab(m+n) = 0$. Since R is a domain, $ab \neq 0$. Also $a(rm) = r(am) = 0$ and $a(-m) = 0$. Zero is torsion, so these elements form a submodule.
Check your argument. The domain hypothesis ensures the product annihilator is nonzero.

MM7033-U3.10 · A · 50 min
Learning objective: U3.1 Maps from a quotient
Prove naturally that $\mathrm{Hom}_{R(\frac{R}{I},M)}$ is isomorphic to $\{m \in M : Im = 0\}$ as an R-module.
Decisive idea. Reconstruct by $[r] \mapsto rm$.
Evaluation at $[1]$ sends a linear map to an element annihilated by I, since $i[1] = 0$. Conversely, if $Im = 0$, the formula $[r] \mapsto rm$ is representative-independent and linear. The two constructions are inverse and preserve addition and scalars. Postcomposition with a module map commutes with evaluation, giving naturality in M.
Check your argument. Check well-definedness and the naturality statement.

MM7033-U3.11 · A · 50 min
Learning objective: U3.2 Splitting a torsion module
Prove $\frac{\mathbb{Z}}{12}\mathbb{Z} \cong \frac{\mathbb{Z}}{3}\mathbb{Z} \oplus \frac{\mathbb{Z}}{4}\mathbb{Z}$ as modules, and identify the images of a generator.
Decisive idea. Coprimality gives surjectivity by Bézout.
The map $[a]_{\{12\}} \mapsto ([a]_3, [a]_4)$ is well defined with trivial kernel, because an integer divisible by three and four is divisible by twelve. Both sides have twelve elements, so it is onto; alternatively $4r - 3s$ lifts $([r]_3, [s]_4)$. The generator $[1]$ maps to $([1], [1])$.
Check your argument. Check the lifting formula modulo each denominator.

MM7033-U3.12 · A · 50 min
Learning objective: U3.1 Submodule invariance
For a vector space with operator T, prove its $K[x]$-submodules are exactly its T-invariant vector subspaces.
Decisive idea. Closure under T gives closure under every polynomial in T.
A $K[x]$-submodule is a K-subspace and is closed under multiplication by x, hence under T. Conversely a K-subspace stable under T is stable under every power of T by induction, and under every finite linear combination of these powers. Thus it is stable under all $f(T)$ and is a submodule.
Check your argument. Prove both implications and explain the polynomial action.

4 Unit 4 · Free modules and exact sequences

MM7033-U4.01 · E · 15 min
Learning objective: U4.1 Presentations

Give a one-generator, one-relation presentation of $\frac{\mathbb{Z}}{9}\mathbb{Z}$.

Decisive idea. Its image is exactly the kernel of reduction.

The presentation is $\mathbb{Z} \xrightarrow{\times 9} \mathbb{Z} \to \frac{\mathbb{Z}}{9}\mathbb{Z} \to 0$. Reduction is onto and its kernel is $9\mathbb{Z}$, the image of the first map.

Check your argument. Verify exactness at the middle and right terms.

MM7033-U4.02 · E · 15 min
Learning objective: U4.2 Exactness

Check exactness of $0 \to K \xrightarrow{i} K^2 \xrightarrow{p} K \to 0$, where $i(a) = (a, 0)$ and $p(a, b) = b$.

Decisive idea. Both the kernel of p and image of i are the first coordinate axis.

i is injective, p is onto, and $\ker p = \{(a, 0)\} = \mathrm{im} i$. Thus the sequence is short exact. A section is $b \mapsto (0, b)$.

Check your argument. Check all three requirements.

MM7033-U4.03 · E · 15 min
Learning objective: U4.1 Finite support

Give an element of $\prod_{n\geq 0} \mathbb{Z}$ not lying in $\bigoplus_{n\geq 0} \mathbb{Z}$.

Decisive idea. The all-ones sequence works.

The tuple $(1, 1, 1, \ldots)$ belongs to the product because every coordinate is an integer. Its support is infinite, so it is absent from the direct sum.

Check your argument. State the support condition.

MM7033-U4.04 · E · 15 min
Learning objective: U4.1 Free maps

How is a linear map $R^3 \to M$ determined?

Decisive idea. Extend by scalar combinations.

An arbitrary triple $m_1, m_2, m_3 \in M$ gives the unique map $(r_1, r_2, r_3) \mapsto r_1 m_1 + r_2 m_2 + r_3 m_3$. Conversely every linear map has this form.

Check your argument. Explain uniqueness by basis expansion.

MM7033-U4.05 · C · 30 min
Learning objective: U4.2 Nonsplitting

Prove $0 \to \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \to \frac{\mathbb{Z}}{3}\mathbb{Z} \to 0$ does not split.

Decisive idea. No nonzero integer is annihilated by three.

If s were a section, then $3s([1]) = s(3[1]) = 0$, so $s([1]) = 0$. Reducing this image modulo three gives zero, contradicting that the composite is the identity on $[1]$.

Check your argument. Use linearity and the section equation.

MM7033-U4.06 · C · 30 min
Learning objective: U4.2 Free quotients

Show a short exact sequence with quotient R^n splits.

Decisive idea. Extend the chosen lifts linearly.

Choose b_i mapping to the ith standard basis vector under the surjection. Define $s(r_1, ..., r_n) = \sum_i r_i b_i$. This is linear and composing with the surjection returns the input vector. Thus it is a section and the splitting lemma applies.

Check your argument. Construct an actual section.

MM7033-U4.07 · C · 30 min
Learning objective: U4.1 Presentation computation

Identify the $\mathbb{Z}$-module generated by u, v with relations $u - 2v = 0$ and $6v = 0$.

Decisive idea. The remaining generator has order six.

Every element becomes a multiple of v, and the remaining relation is $6v = 0$. The map to $\frac{\mathbb{Z}}{6}\mathbb{Z}$ sending v to $[1]$ and u to $[2]$ respects both relations. The reverse map sends $[1]$ to v, and their composites are identities. Hence the module is $\frac{\mathbb{Z}}{6}\mathbb{Z}$.

Check your argument. Prove no extra hidden relation remains by constructing inverse maps.

MM7033-U4.08 · C · 30 min
Learning objective: U4.2 Direct summands

Prove a direct summand of a projective module is projective.

Decisive idea. Lift through the larger module and restrict back.

Write $P = A \oplus B$ with P projective. Given $f : A \to N$ and a surjection $M \to N$, compose f with the projection $P \to A$, lift that map to M using projectivity, then restrict along $A \to P$. The resulting map lifts f. Thus A is projective.

Check your argument. State the extension and restriction maps.

MM7033-U4.09 · C · 30 min
Learning objective: U4.1 Finite rank
Prove R^2 cannot be isomorphic to R^3 for a nonzero commutative unital ring.
Decisive idea. An isomorphism induces an isomorphism of residue-field vector spaces.
Choose a maximal ideal M. Quotienting the two modules by M times themselves would give an isomorphism $\left(\frac{R}{M}\right)^2 \cong \left(\frac{R}{M}\right)^3$. Vector-space dimension rules this out.
Check your argument. Explain why the module isomorphism descends to these quotients.

MM7033-U4.10 · A · 50 min
Learning objective: U4.2 Projective cyclic example
Show $\frac{\mathbb{Z}}{2}\mathbb{Z}$ is not a projective $\mathbb{Z}$-module.
Decisive idea. Lifting the identity would provide a section.
If the module were projective, its identity would lift through the surjection from $\mathbb{Z}$, giving a section. Such a section must send $[1]$ to an integer killed by two, hence to zero, contradicting the section condition. Therefore it is not projective.
Check your argument. Use the definition of projectivity with fully specified maps.

MM7033-U4.11 · A · 50 min
Learning objective: U4.2 A projective nonfree module
Let $R = K \times K$ and $e = (1, 0)$. Prove $\mathfrak{R}$ is projective but is not free.
Decisive idea. Every element of $\mathfrak{R}$ is annihilated by $(0, 1)$.
The decomposition $R = Re \oplus R(1 - e)$ follows by splitting $(a, b) = (a, 0) + (0, b)$. It makes Re a direct summand of the free module R, hence projective. It is nonzero and is annihilated by the nonzero element $(0, 1)$. A nonzero free module has a basis vector not annihilated by any nonzero scalar, since its corresponding coordinate is that scalar. Thus $\mathfrak{R}$ is not free of any rank.
Check your argument. Exclude rank zero and justify the annihilator property of free modules.

MM7033-U4.12 · A · 50 min
Learning objective: U4.2 Splitting from a retraction
Given a short exact sequence and a retraction $r : B \to A$ of its injection i, construct the isomorphism $B \cong A \oplus C$.
Decisive idea. The restriction of the quotient map to $\ker r$ is an isomorphism.
The map $p : \ker r \to C$ is injective because a kernel element is $i(a)$ with $a = ri(a) = 0$. It is onto because any lift b of c can be replaced by $b - i(r(b))$. Let s be its inverse. Then $(a, c) \mapsto i(a) + s(c)$ has inverse $b \mapsto (r(b), p(b))$. Direct substitution verifies both identities.
Check your argument. Prove the restriction is bijective before using its inverse.

5 Unit 5 · Tensor products

MM7033-U5.01 · E · 15 min
Learning objective: U5.1 Tensor relations
Simplify $(2m) \otimes n + m \otimes (3n)$ over $\mathbb{Z}$.
Decisive idea. Both terms are multiples of $m \otimes n$.
The expression is $2(m \otimes n) + 3(m \otimes n) = 5(m \otimes n)$ by bilinearity.
Check your argument. Use tensor relations, not an interpretation as ordered pairs.

MM7033-U5.02 · E · 15 min
Learning objective: U5.2 Cyclic tensors
Compute $\frac{\mathbb{Z}}{8}\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{12}\mathbb{Z}$.
Decisive idea. The gcd is four.
The tensor is $\frac{\mathbb{Z}}{4}\mathbb{Z}$, since tensoring the second module with $\frac{\mathbb{Z}}{8}\mathbb{Z}$ gives its quotient by eight times itself, equivalently $\frac{\mathbb{Z}}{8\mathbb{Z}+12\mathbb{Z}} = \frac{\mathbb{Z}}{4}\mathbb{Z}$.
Check your argument. Explain the quotient computation.

MM7033-U5.03 · E · 15 min
Learning objective: U5.1 Unit tensor
Give inverse maps proving $R \otimes_R M \cong M$.
Decisive idea. Use $m \mapsto 1 \otimes m$ for the inverse.
The maps are $r \otimes m \mapsto rm$ and $m \mapsto 1 \otimes m$. Their composites are identities because $1m = m$ and $1 \otimes rm = r \otimes m$ by scalar movement.
Check your argument. Check well-definedness by bilinearity.

MM7033-U5.04 · E · 15 min
Learning objective: U5.2 Tensor annihilation
Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{7}\mathbb{Z}$.
Decisive idea. Multiplication by seven is onto on $\mathbb{Q}$.
The tensor is $\frac{\mathbb{Q}}{7}\mathbb{Q} = 0$, since any rational q is seven times $\frac{q}{7}$.
Check your argument. Distinguish this from the Cartesian product of the two groups.

MM7033-U5.05 · C · 30 min
Learning objective: U5.1 Pure tensors

Show $e_1 \otimes e_1 + e_2 \otimes e_2$ in $K^2 \otimes K^2$ is not a pure tensor.

Decisive idea. A pure tensor gives a matrix of rank at most one.

Write $u = (a, b), v = (c, d)$. Its coefficient matrix is $\begin{pmatrix} ac & ad \\ bc & bd \end{pmatrix}$, whose determinant is zero. The displayed tensor has coefficient matrix the identity, determinant one. The coefficients are uniquely determined by the tensor basis, so no pure tensor can equal it.

Check your argument. Justify the basis-coordinate comparison.

MM7033-U5.06 · C · 30 min
Learning objective: U5.1 Tensor basis

Prove $R^m \otimes_R R^n$ is free of rank mn.

Decisive idea. Coordinate-product bilinear maps detect each coefficient.

The tensors $e_i \otimes f_j$ span by bilinearity. For each pair (i, j), the bilinear map $(u, v) \mapsto u_i v_j$ induces a linear functional taking value one on its matching tensor and zero on the others. These functionals prove independence. Thus the displayed mn tensors form a basis.

Check your argument. Prove independence over a ring, not only over a field.

MM7033-U5.07 · C · 30 min
Learning objective: U5.2 Quotient tensors

Compute $\frac{R}{I} \otimes_R \frac{R}{J}$.

Decisive idea. The resulting quotient kills both ideals.

The tensor is $\frac{\frac{R}{J}}{I(\frac{R}{J})}$, canonically isomorphic to $\frac{R}{I+J}$. The map sends $[a] \otimes [b]$ to $ab + (I + J)$, and its inverse sends $r + (I + J)$ to $[1] \otimes [r]$.

Check your argument. State the canonical maps or justify the quotient identifications.

MM7033-U5.08 · C · 30 min
Learning objective: U5.2 Failure of injectivity

Tensor $0 \to \mathbb{Z} \xrightarrow{\times 4} \mathbb{Z} \to \frac{\mathbb{Z}}{4}\mathbb{Z} \to 0$ with $\frac{\mathbb{Z}}{2}\mathbb{Z}$. Describe every resulting map.

Decisive idea. The first induced map is multiplication by four modulo two.

The nonzero portion is $\frac{\mathbb{Z}}{2}\mathbb{Z} \xrightarrow{0} \frac{\mathbb{Z}}{2}\mathbb{Z} \xrightarrow{\mathrm{id}} \frac{\mathbb{Z}}{2}\mathbb{Z} \to 0$. The first map is zero; the quotient map induces the identity on the generator tensors. The sequence remains exact at the last two terms but not injective at the first.

Check your argument. Explain why a leading zero cannot be retained as an exact term.

MM7033-U5.09 · C · 30 min
Learning objective: U5.1 Distributivity

Prove $(M \oplus N) \otimes P \cong (M \otimes P) \oplus (N \otimes P)$.

Decisive idea. Reverse the map using the two inclusions.

Define $(m, n) \otimes p \mapsto (m \otimes p, n \otimes p)$, which is bilinear and hence descends. In the reverse direction send $m \otimes p$ to $(m, 0) \otimes p$ and $n \otimes p$ to $(0, n) \otimes p$, then add. Their composites are identities on pure tensors and hence everywhere.

Check your argument. Verify that the formulas respect the universal properties.

MM7033-U5.10 · A · 50 min
Learning objective: U5.2 Right exactness

For a submodule $K \leq M$, identify the kernel of $M \otimes N \to \left(\frac{M}{K}\right) \otimes N$.

Decisive idea. Do not assume the map $K \otimes N \to M \otimes N$ is injective.

The kernel is the image of $K \otimes N$, equivalently the submodule generated by all $k \otimes n$ with $k \in K$. Quotienting by this submodule classifies bilinear maps on $M \times N$ vanishing on $K \times N$, exactly the maps descending to $\left(\frac{M}{K}\right) \times N$. Thus this quotient is the target tensor. The source map from $K \otimes N$ may have its own kernel.

Check your argument. Give the universal-property argument and distinguish image from an embedded copy.

MM7033-U5.11 · A · 50 min
Learning objective: U5.1 Tensor–Hom correspondence

Prove $\mathrm{Hom}_{R(M \otimes N, P)} \cong \mathrm{Hom}_{R\left(M, \mathrm{Hom}_{R(N,P)}\right)}$ naturally.

Decisive idea. A map $m \mapsto (n \mapsto b(m, n))$ is linear exactly when b is bilinear.

A linear map from the tensor yields a bilinear $b(m, n)$, which curries to a linear map into the pointwise module of linear maps $N \to P$. Conversely evaluate such a curried map on n to obtain a bilinear map, then use the tensor universal property. Currying and evaluation are inverse, and pre- or postcomposition commutes with both formulas, proving naturality.

Check your argument. Check linearity in each variable and explain the module structure on Hom.

MM7033-U5.12 · A · 50 min
Learning objective: U5.2 A torsion tensor vanishes

Prove $\left(\frac{\mathbb{Q}}{\mathbb{Z}}\right) \otimes_{\mathbb{Z}\left(\frac{\mathbb{Q}}{\mathbb{Z}}\right)} = 0$.

Decisive idea. The second factor is divisible by that same integer.

For a pure tensor $x \otimes y$, choose positive n with $nx = 0$. Writing $y = q + \mathbb{Z}$, take $z = \frac{q}{n} + \mathbb{Z}$ so that $nz = y$. This proves the divisibility used here directly. Then $x \otimes y = x \otimes nz = nx \otimes z = 0$. Every tensor is a finite sum of pure tensors, so the whole tensor product vanishes.

Check your argument. Explain both torsion and divisibility and conclude for sums, not only pure tensors.

6 Unit 6 · Localization

MM7033-U6.01 · E · 15 min
Learning objective: U6.1 Localized integers
Describe $\mathbb{Z}\left[\frac{1}{3}\right]$ as a subring of $\mathbb{Q}$.
Decisive idea. Fractions may simplify but need not have a unique presentation.
It is $\{\frac{a}{3^n} : a \in \mathbb{Z}, n \geq 0\}$. Addition and multiplication keep this form. It is the smallest subring of $\mathbb{Q}$ containing $\mathbb{Z}$ and $\frac{1}{3}$.
Check your argument. Specify allowed denominators.

MM7033-U6.02 · E · 15 min
Learning objective: U6.1 Units at a prime
Which of $2, 5, 10$ are units in $\mathbb{Z}_{(5)}$?
Decisive idea. An integer is a unit exactly when five does not divide it.
Two is a unit, with inverse $\frac{1}{2}$. Five and ten lie in the unique maximal ideal and are not units. If either had an inverse with denominator not divisible by five, cross multiplication would force that denominator divisible by five.
Check your argument. Justify the nonunit claims.

MM7033-U6.03 · E · 15 min
Learning objective: U6.1 Vanishing
In $\frac{\mathbb{Z}}{10}\mathbb{Z}$, localize at powers of two. Show the class of five becomes zero.
Decisive idea. Two times five is zero in the original ring.
The element two belongs to the denominator set and annihilates five. Hence $\frac{[5]}{1} = 0$ in the localization by the defining fraction relation.
Check your argument. Do not cancel two inside the original ring.

MM7033-U6.04 · E · 15 min
Learning objective: U6.2 Localizing a cyclic module
Compute $\mathbb{Z}\left[\frac{1}{2}\right] \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{8}\mathbb{Z}$.
Decisive idea. Eight is now a unit.
The tensor is $\frac{\mathbb{Z}\left[\frac{1}{2}\right]}{8}\mathbb{Z}\left[\frac{1}{2}\right] = 0$, since multiplication by eight is invertible in the localized ring.
Check your argument. Explain why the quotient is zero.

MM7033-U6.05 · C · 30 min
Learning objective: U6.1 Injectivity criterion
Prove the map $R \to S^{-1}R$ is injective exactly when no element of S annihilates a nonzero element of R.
Decisive idea. An element maps to zero iff some denominator-set element kills it.
The kernel is $\{a : ua = 0 \text{ for some } u \in S\}$. It is zero precisely under the stated condition. This equivalence also handles the case $0 \in S$, where injectivity fails unless R is zero.
Check your argument. State both directions and the zero-ring edge case.

MM7033-U6.06 · C · 30 min
Learning objective: U6.1 Identifying a localization
Prove localizing $\frac{\mathbb{Z}}{6}\mathbb{Z}$ at powers of two gives $\mathbb{F}_3$.
Decisive idea. The kernel that disappears is generated by three.
Reduction gives a map to $\mathbb{F}_3$ sending every power of two to a unit, so it extends to the localization. In that localization three is zero because two kills it. Thus the original map factors through $\frac{\mathbb{Z}}{3}\mathbb{Z}$, and two is already invertible there. The resulting maps in both directions agree on original elements and inverses of two, so they are inverse.
Check your argument. Use the universal property to verify the inverse maps.

MM7033-U6.07 · C · 30 min
Learning objective: U6.2 A common annihilator
If M is finitely generated and $S^{-1}M = 0$, prove some $s \in S$ kills all of M.
Decisive idea. Multiply the finitely many annihilators.
For generators $m_1, \ldots, m_n$, the vanishing criterion gives $s_i \in S$ with $s_i m_i = 0$. The product $s = \prod_i s_i$ belongs to S and kills every generator, hence every linear combination. Thus $sM = 0$. For the zero module choose $s = 1$.
Check your argument. Use finite generation explicitly.

MM7033-U6.08 · C · 30 min
Learning objective: U6.2 Quotients and localization
Show $S^{-1}\left(\frac{M}{N}\right) \cong S^{-1}\frac{M}{\operatorname{im}(S^{-1}N)}$.
Decisive idea. Exactness identifies the cokernel.
Localizing $N \to M \to \frac{M}{N} \to 0$ remains exact, so the target is the quotient of $S^{-1}M$ by the image of the localized submodule. Explicitly $\frac{m}{s}$ maps to $\frac{m+N}{s}$. Since localization preserves injections, that image can also be identified with $S^{-1}N$.
Check your argument. Explain both the image formulation and the injective identification.

MM7033-U6.09 · C · 30 min
Learning objective: U6.1 Local rings
Show the nonunits in $R_{\mathfrak{p}}$ form its unique maximal ideal.
Decisive idea. Fractions with numerator inside it form a proper ideal.
The numerator-in-$\mathfrak{p}$ fractions form a proper ideal by the prime-localization theorem. Every fraction outside it has an invertible numerator and is a unit. No element of a proper ideal can be a unit. Thus this ideal is exactly the nonunits and contains every proper ideal, making it uniquely maximal.
Check your argument. Use properness as well as the unit calculation.

MM7033-U6.10 · A · 50 min
Learning objective: U6.2 Detecting zero locally
Prove an R-module M is zero if $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$. No finite-generation hypothesis is allowed.
Decisive idea. Choose a maximal ideal containing that annihilator.
If $m \in M$ is nonzero, its annihilator is proper and lies in some maximal ideal $\mathfrak{m}$. If $\frac{m}{1} = 0$ in that localization, some $s \notin \mathfrak{m}$ kills m, placing s in the annihilator and hence in $\mathfrak{m}$, a contradiction. Thus some localization is nonzero. The contrapositive proves the claim.
Check your argument. Work with one element's annihilator; do not assume a finite generator list.

MM7033-U6.11 · A · 50 min
Learning objective: U6.1 Localization in stages
Show inverting f and then g in a commutative ring is equivalent to inverting fg.
Decisive idea. If fg is a unit, both f and g are units.
A map out of the iterated localization is exactly a map from R sending both f, g to units. This is equivalent to sending fg to a unit: if u inverts fg, then gu inverts f and fu inverts g, using commutativity. Thus the two rings have the same universal property and are canonically isomorphic.
Check your argument. Prove the implication from the product being a unit.

MM7033-U6.12 · A · 50 min
Learning objective: U6.2 Flatness of localization
Prove $S^{-1}R$ is flat by explicitly checking that an injection remains injective after localization and identifying localization with tensoring.
Decisive idea. An element of S killing the image must kill the original element by injectivity.
For injective $f : M \to N$, if $\frac{f(m)}{s} = 0$, there is $u \in S$ with $f(um) = 0$. Injectivity gives $um = 0$, so $\frac{m}{s} = 0$. Hence localization preserves injections. The natural isomorphism $S^{-1}R \otimes M \cong S^{-1}M$ transfers this property to tensoring with $S^{-1}R$, which is flatness.
Check your argument. State the tensor identification and the exact vanishing criterion.

7 Unit 7 · Multilinear algebra

MM7033-U7.01 · E · 15 min
Learning objective: U7.1 Exterior arithmetic

Expand $(e_1 + 2e_2) \wedge (3e_1 + e_2)$ in $\wedge^2 R^2$.

Decisive idea. Use $e_2 \wedge e_1 = -e_1 \wedge e_2$.

The expansion is $e_1 \wedge e_2 + 6e_2 \wedge e_1 = -5e_1 \wedge e_2$. The coefficient is interpreted in R, so it can vanish in characteristic five.

Check your argument. Keep the sign and coefficient-ring interpretation.

MM7033-U7.02 · E · 15 min
Learning objective: U7.1 Exterior rank

Give a basis and rank of $\wedge^2 R^4$.

Decisive idea. There are six pairs.

A basis is $e_1 \wedge e_2, e_1 \wedge e_3, e_1 \wedge e_4, e_2 \wedge e_3, e_2 \wedge e_4, e_3 \wedge e_4$. Its rank is $\binom{4}{2} = 6$.

Check your argument. Use increasing indices once each.

MM7033-U7.03 · E · 15 min
Learning objective: U7.2 Trace and determinant

Find trace and determinant of $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ over $\mathbb{Z}$.

Decisive idea. The off-diagonal term contributes no determinant product here.

The trace is five and determinant six. Equivalently the operator acts on the top wedge by multiplying $e_1 \wedge e_2$ by six.

Check your argument. Connect at least one computation to the intrinsic definition.

MM7033-U7.04 · E · 15 min
Learning objective: U7.1 Repeated inputs

Why is $v \wedge v = 0$ even in characteristic two?

Decisive idea. It is an imposed alternating relation.

The exterior power is defined by killing tensors with repeated inputs, so $v \wedge v = 0$ in every characteristic. In characteristic two it does not follow merely from the equation $v \wedge v = -v \wedge v$, which would only say twice it is zero.

Check your argument. Distinguish alternating from a sign-only condition.

MM7033-U7.05 · C · 30 min
Learning objective: U7.2 Cyclic trace
Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for square matrices over a commutative ring.
Decisive idea. Interchange the two indices and commute scalar factors.
$\operatorname{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji}$. Commutativity of the ring changes each term to $b_{ji} a_{ij}$, and interchanging indices gives $\operatorname{tr}(BA)$. No division or field assumption is required.
Check your argument. Identify where commutativity enters.

MM7033-U7.06 · C · 30 min
Learning objective: U7.2 Similarity invariance
Prove similar matrices have equal determinant and trace.
Decisive idea. Use determinant multiplicativity and cyclicity of trace.
$\det(P^{-1}AP) = \det(P)^{-1} \det(A) \det(P) = \det(A)$. Also $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A)$. The inverse determinant relation follows by taking determinants of $PP^{-1} = 1$.
Check your argument. Justify the inverse identities.

MM7033-U7.07 · C · 30 min
Learning objective: U7.1 Functorial exterior powers
Show $\wedge^{n(TS)} = (\wedge^n T)(\wedge^n S)$.
Decisive idea. Both sides apply TS to each input.
On $m_1 \wedge \ldots \wedge m_n$, both sides give $TSm_1 \wedge \ldots \wedge TSm_n$. Such wedges generate the exterior power, and both maps are linear, so equality holds everywhere.
Check your argument. Explain why checking generators suffices.

MM7033-U7.08 · C · 30 min
Learning objective: U7.2 Rank-one endomorphisms
Let $T : M \to M$ have form $T(m) = \varphi(m)v$ on a finite free module. Prove $\operatorname{tr}(T) = \varphi(v)$.
Decisive idea. Expand v in that basis.
For basis e_i with coordinate functionals ε_i, the trace is $\sum_i \varepsilon_{i(Te_i)} = \sum_i \varphi(e_i)\varepsilon_{i(v)}$. Linearity of φ and $v = \sum_i \varepsilon_{i(v)} e_i$ identify this with $\varphi(v)$.
Check your argument. Show that the result does not depend on the chosen basis.

MM7033-U7.09 · C · 30 min
Learning objective: U7.1 Vanishing top powers

Prove $\wedge^n R^r = 0$ for $n > r$.

Decisive idea. Every n-tuple of basis indices repeats an index.

Multilinearity expresses each wedge as a linear combination of wedges of standard basis elements. With more slots than available indices, each term repeats an index and is zero by alternation. Hence every generator and thus every element vanishes.

Check your argument. Use the pigeonhole step explicitly.

MM7033-U7.10 · A · 50 min
Learning objective: U7.2 Dual bases

For a finite projective module, prove the trace formula is independent of the chosen dual basis.

Decisive idea. Use the reconstruction identity to collapse the double sum.

For dual families (m_i, φ_i) and (n_j, ψ_j), expand $m_i = \sum_j \psi_{j(m_i)} n_j$. The first trace becomes $\sum_{i,j} \psi_{j(m_i)} \varphi_{i(Tn_j)}$. For fixed j, the reconstruction $Tn_j = \sum_i \varphi_{i(Tn_j)} m_i$ makes the inner sum equal $\psi_{j(Tn_j)}$. Summing gives the second trace formula.

Check your argument. Use only dual-basis identities, not an assumption that either family is linearly independent.

MM7033-U7.11 · A · 50 min
Learning objective: U7.1 Exterior direct-sum decomposition

For finite free modules M, N, prove $\wedge^2 (M \oplus N) \cong \wedge^2 M \oplus (M \otimes N) \oplus \wedge^2 N$.

Decisive idea. Sorted pairs are of three types: both in M, one in each, or both in N.

Choose bases e_i, f_j for the two summands. The exterior basis on their concatenation splits into $e_i \wedge e_k$ with $i < k$, $e_i \wedge f_j$, and $f_j \wedge f_l$ with $j < l$. These match bases of the three displayed modules. The map is induced by their inclusions, with the middle term $m \otimes n \mapsto (m, 0) \wedge (0, n)$. Thus it is independent of the chosen bases, whose role was to prove bijectivity.

Check your argument. Define the canonical map and use the basis decomposition to establish it is an isomorphism.

MM7033-U7.12 · A · 50 min
Learning objective: U7.2 Finite-order operators

Let $T^5 = 1$ on a finite-dimensional vector space. Show $\det(T)^5 = 1$, and give a reason this does not force T to be the identity.

Decisive idea. A nontrivial fifth-root scalar operator is a counterexample over $\mathbb{C}$.

Multiplicativity gives $\det(T)^5 = \det(T^5) = 1$. Over $\mathbb{C}$, multiplication by a primitive fifth root of unity on a one-dimensional space is nonidentity but its fifth power is identity. Even determinant one would not force a higher-dimensional operator to be identity, for example a diagonal matrix with reciprocal nontrivial eigenvalues.

Check your argument. Separate the scalar determinant constraint from the full operator.

8 Unit 8 · Noetherian methods and applications

MM7033-U8.01 · E · 15 min
Learning objective: U8.1 Noetherian examples

Explain why every finite-dimensional vector space is a Noetherian module over its field.

Decisive idea. Equal-dimensional nested subspaces are equal.

The dimensions in an increasing chain are nondecreasing integers bounded by the space's finite dimension. They eventually stabilize, and nested subspaces of equal finite dimension are equal. Thus the chain stabilizes.

Check your argument. Justify equality once dimensions stop increasing.

MM7033-U8.02 · E · 15 min
Learning objective: U8.1 Hilbert's theorem

Why is $K[x, y, z]$ Noetherian for a field K?

Decisive idea. Apply Hilbert's theorem three times.

A field is Noetherian because its ideals are zero and the whole field. Hilbert's theorem successively gives Noetherianity of $K[x]$, then $K[x, y]$, then $K[x, y, z]$.

Check your argument. Name the base case and each finite-variable extension.

MM7033-U8.03 · E · 15 min
Learning objective: U8.2 Coordinate rings

Show $\frac{K[x,y]}{xy}$ is not a domain.

Decisive idea. Neither class lies in the ideal, but their product does.

The classes of x and y are nonzero: a multiple of xy cannot equal either polynomial, as evaluation at $y = 0$ or $x = 0$ confirms. Their product is zero by the defining relation. Hence the quotient has zero divisors.

Check your argument. Prove the two factors are nonzero.

MM7033-U8.04 · E · 15 min
Learning objective: U8.2 Specialization

Compute $R^3 \otimes_R \frac{R}{M}$ for a maximal ideal M.

Decisive idea. Each copy of R becomes $\frac{R}{M}$.

The result is $\left(\frac{R}{M}\right)^3$, a three-dimensional vector space over the residue field. The standard basis tensors give its basis.

Check your argument. Identify the coefficient field.

MM7033-U8.05 · C · 30 min
Learning objective: U8.1 Quotients

Prove a quotient of a Noetherian module is Noetherian.

Decisive idea. The preimages form an increasing chain of submodules.

An increasing chain of quotient submodules lifts to a chain of submodules containing the kernel. The original module is Noetherian, so the lifted chain stabilizes. Taking images under the quotient map shows the original quotient chain stabilizes.

Check your argument. Use the submodule correspondence.

MM7033-U8.06 · C · 30 min
Learning objective: U8.1 Finitely generated submodules

Prove the ascending chain condition implies every submodule is finitely generated.

Decisive idea. At each finite stage choose an element outside the current span.

A non-finitely-generated submodule allows a choice m_1, then m_2 outside Rm_1, then m_3 outside their span, and so on. The resulting finitely generated spans form a strictly increasing chain, contradicting the ascending chain condition. Thus every submodule is finitely generated.

Check your argument. Explain why the process cannot terminate under the contrary assumption.

MM7033-U8.07 · C · 30 min
Learning objective: U8.1 Infinite polynomial ring

Show $K[x_1, x_2, ...]$ is not Noetherian.

Decisive idea. The next variable does not belong to the preceding ideal.

The chain $(x_1) \subset (x_1, x_2) \subset ...$ is strictly increasing. Indeed setting the first n variables to zero kills the nth ideal but does not kill x_{n+1}. Thus the chain never stabilizes.

Check your argument. Prove strictness at every stage.

MM7033-U8.08 · C · 30 min
Learning objective: U8.2 Local coordinate calculation

Compute the localization at x of $A = \frac{K[x,y]}{xy}$.

Decisive idea. The remaining coordinate is an invertible x.

The localization is $K\left[x, x^{-1}\right]$. A map sends the class of x to x, its inverse to x^{-1}, and y to zero. Conversely the invertible class of x defines a map from the Laurent polynomial ring into A_x. The composites fix all generators, including inverses, so are identities.

Check your argument. Give both maps to rule out extra relations.

MM7033-U8.09 · C · 30 min
Learning objective: U8.2 Finite presentation

Prove every finitely generated module over a Noetherian ring is finitely presented.

Decisive idea. Its kernel is finitely generated.

Choose $R^n \to M$ onto. The finite free module is Noetherian, so its kernel has finitely many generators. A map from R^m taking basis elements to those generators gives an exact presentation $R^m \to R^n \to M \to 0$.

Check your argument. Identify where Noetherianity is used.

MM7033-U8.10 · A · 50 min
Learning objective: U8.1 Localization preserves Noetherianity

Prove a localization of a Noetherian ring is Noetherian.

Decisive idea. Every fraction in the ideal has its numerator in the contracted ideal after multiplying by a unit.

Let J be an ideal of $S^{-1}R$ and $I = \{a : \frac{a}{1} \in J\}$. Since R is Noetherian, $I = (a_1, ..., a_n)$. If $\frac{a}{s} \in J$, multiplying by $\frac{s}{1}$ gives $\frac{a}{1} \in J$, so $a \in I$. Thus $\frac{a}{s}$ is a combination of $\frac{a_i}{1}$ with coefficients in the localization. Conversely these generators lie in J, so they generate it. Every ideal is finitely generated.

Check your argument. Prove the contraction–extension equality used in the argument.

MM7033-U8.11 · A · 50 min
Learning objective: U8.2 Eigenspace decomposition

Suppose $T^2 = 1$ on a vector space over a field of characteristic not two. Construct projections onto the two eigenspaces and prove a direct-sum decomposition.

Decisive idea. Take $\frac{1+T}{2}$ and $\frac{1-T}{2}$.

Let $P_+ = \frac{1+T}{2}$ and $P_- = \frac{1-T}{2}$. Using $T^2 = 1$ gives $P_+^2 = P_+$, $P_-^2 = P_-$, $P_+P_- = 0$, and $P_+ + P_- = 1$. Also $TP_+ = P_+$ and $TP_- = -P_-$. Hence their images lie in the two eigenspaces and every vector is their sum. The intersection is zero because the projections are orthogonal, giving the direct sum.

Check your argument. The inverse of two is essential; state the characteristic hypothesis.

MM7033-U8.12 · A · 50 min
Learning objective: U8.1 Hilbert reduction step

In the proof of Hilbert's theorem, explain in detail why finitely many polynomials with leading coefficients generating the leading-coefficient ideal suffice together with finitely many bounded-degree polynomials.

Decisive idea. The bounded-degree intersection is a submodule of a finite free module.

Let f_i have degrees d_i and put $N = \max d_i$. For any ideal element of degree $d \geq N$, express its leading coefficient as $\sum r_i \operatorname{lc}(f_i)$ and subtract $\sum r_i x^{d-d_i} f_i$ to lower the degree. Repeating terminates below N. The ideal's intersection with polynomials of degree less than N is finitely generated as an R-module because R^N is Noetherian. Those generators finish the reduction.

Check your argument. Explain termination and why coefficient generation alone does not handle low degrees.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7033-M1.01 · E · 15 min
Learning objective: U1 Finite fields

List the subfield sizes of $\mathbb{F}_{2^8}$ and justify that the list is complete.

Decisive idea. The allowed degrees are one, two, four, and eight.

The sizes are $2, 4, 16, 256$, one subfield of each size. Necessity follows from the tower law and sufficiency and uniqueness from the finite-field subfield theorem.

Check your argument. State all three aspects: necessity, existence, uniqueness.

MM7033-M1.02 · E · 15 min
Learning objective: U5 Tensor computation

Compute $\frac{\mathbb{Z}}{15}\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{21}\mathbb{Z}$.

Decisive idea. Their gcd is three.

The tensor is $\frac{\mathbb{Z}}{15\mathbb{Z}+21\mathbb{Z}} = \frac{\mathbb{Z}}{3}\mathbb{Z}$, by the cyclic tensor formula and Bézout.

Check your argument. Name the canonical generator tensor.

MM7033-M1.03 · C · 30 min
Learning objective: U3–U4 Presentations

Identify the module generated by a, b with relations $a = 3b$ and $10b = 0$, and decide whether it is projective over $\mathbb{Z}$.

Decisive idea. A section of $\mathbb{Z} \to \frac{\mathbb{Z}}{10}\mathbb{Z}$ cannot exist.

The module is $\frac{\mathbb{Z}}{10}\mathbb{Z}$ via $b \mapsto [1]$, $a \mapsto [3]$, with the inverse sending $[1]$ to b. It is not projective: lifting its identity across the quotient map would give a section, but the image of $[1]$ would be an integer killed by ten and hence zero, contradicting the section equation.

Check your argument. Give the isomorphism and a separate projectivity argument.

MM7033-M1.04 · C · 30 min
Learning objective: U6 Localization

Compute $\mathbb{Z}\left[\frac{1}{3}\right] \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{12}\mathbb{Z}$.

Decisive idea. Three is a unit, so the ideal generated by twelve equals that generated by four.

The tensor is $\frac{\mathbb{Z}\left[\frac{1}{3}\right]}{12}\mathbb{Z}\left[\frac{1}{3}\right] = \frac{\mathbb{Z}\left[\frac{1}{3}\right]}{4}\mathbb{Z}\left[\frac{1}{3}\right]$. Since three is already invertible modulo four, this quotient is $\frac{\mathbb{Z}}{4}\mathbb{Z}$. The natural reduction map sends $\frac{a}{3^n}$ to $[a][3]^{-n}$.

Check your argument. Explain the last localization identification.

MM7033-M1.05 · A · 50 min
Learning objective: U2–U3 Simple quotients

Prove every nonzero finitely generated module over a commutative unital ring has a maximal proper submodule.

Decisive idea. Finite generation prevents a chain union from becoming the whole module without one member doing so.

The poset of proper submodules is nonempty. For a nonempty chain, its union is a submodule; the empty chain has the proper upper bound zero because M is nonzero. If the union were all of M, finitely many generators would all lie in one member, forcing that member to be M, a contradiction. Thus every chain has a proper upper bound. Zorn gives a maximal proper submodule, whose quotient is simple.

Check your argument. Use finite generation exactly in the properness check.

MM7033-M1.06 · A · 50 min
Learning objective: U7–U8 Operators

Let $T^3 = 1$ over $\mathbb{C}$. Prove the space is a direct sum of the three eigenspaces and describe projections using interpolation.

Decisive idea. Use polynomials equal to one at one root and zero at the others.

For roots λ_i, set $e_{i(t)} = \prod_{j \neq i} \frac{t-\lambda_j}{\lambda_i-\lambda_j}$. Modulo $t^3 - 1$, these are orthogonal idempotents summing to one and satisfy $(t - \lambda_i)e_i = 0$. Substituting T gives projections onto the corresponding eigenspaces, with sum the identity and pairwise zero products. Their images therefore give the direct-sum decomposition.

Check your argument. Explain why polynomial congruences remain true after substituting T.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7033-M2.01 · E · 15 min

Learning objective: U3 Module maps

Compute $\mathrm{Hom}_{\mathbb{Z}\left(\frac{\mathbb{Z}}{6}\mathbb{Z},\frac{\mathbb{Z}}{9}\mathbb{Z}\right)}$ as an abelian group.

Decisive idea. That image must be killed by six.

The allowable images are $[0]$, $[3]$, $[6]$ modulo nine. Under pointwise addition these form a cyclic group of order three, so the Hom group is $\frac{\mathbb{Z}}{3}\mathbb{Z}$.

Check your argument. Identify the group operation as well as the number of maps.

MM7033-M2.02 · E · 15 min

Learning objective: U7 Exterior powers

Compute the rank of $\wedge^3 R^5$ and explain why $\wedge^6 R^5 = 0$.

Decisive idea. Repeated basis indices force a wedge to vanish.

The first rank is $\binom{5}{3} = 10$. In six slots drawn from five basis indices, some index repeats, so every basis expansion term vanishes and the sixth exterior power is zero.

Check your argument. Justify both spanning and the repeated-index argument.

MM7033-M2.03 · C · 30 min

Learning objective: U1 Splitting fields

Find the degree of the splitting field of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

Decisive idea. Show the second is not in the first quadratic field.

The splitting field is $\mathbb{Q}\left(\sqrt{2},\sqrt{3}\right)$. If $\sqrt{3} = a + b\sqrt{2}$, squaring forces $ab = 0$, and either $a^2 = 3$ or $b^2 = \frac{3}{2}$, impossible for rational a, b. Thus each adjoining step has degree two and the total degree is four.

Check your argument. Prove the relative second degree, not merely an upper bound.

MM7033-M2.04 · C · 30 min
Learning objective: U8 Noetherianity

Prove $\frac{K[x,y]}{x^2,xy}$ is Noetherian. Must it be a domain?

Decisive idea. The nonzero class of x has square zero.

Hilbert's theorem makes $K[x,y]$ Noetherian, and its quotient is Noetherian. It is not a domain: the class of x is nonzero because the ideal is generated by degree-two monomials, but its square is zero.

Check your argument. Separate Noetherianity from absence of zero divisors.

MM7033-M2.05 · A · 50 min
Learning objective: U5–U6 Local detection of a map

Prove a module map is an isomorphism if its localization at every maximal ideal is an isomorphism.

Decisive idea. Localization is exact, and zero modules can be detected at maximal ideals.

Exactness identifies localized kernels and cokernels with the kernels and cokernels of the localized maps, so both localize to zero at every maximal ideal. The local zero-detection theorem forces the original kernel and cokernel to be zero. The map is therefore injective and surjective, hence an isomorphism.

Check your argument. No finite-generation assumption is needed; cite the element-annihilator proof of zero detection.

MM7033-M2.06 · A · 50 min
Learning objective: U4–U7 Projective trace

Let M be finite projective and $T(m) = \varphi(m)v$. Prove $\operatorname{tr}(T) = \varphi(v)$ using a dual basis.

Decisive idea. Apply φ to the dual-basis reconstruction of v.

For a dual basis (m_i, ψ_i), $\operatorname{tr}(T) = \sum_i \psi_{i(\varphi(m_i)v)} = \sum_i \varphi(m_i)\psi_{i(v)}$. Since $v = \sum_i \psi_{i(v)} m_i$, applying φ gives exactly this sum. Thus the trace is $\varphi(v)$, without assuming a global ordinary basis.

Check your argument. Use scalar commutativity and the correct dual-basis identity.